6	(a)	<u>magnitude of induced electromotive force</u> <u>proportional</u> to <u>rate of change of magnetic flux linkage</u> **or** rate of magnetic flux cutting		**B1** **B1**	
	(b)				
		(i)	input power/energy = output power/energy **or** no power/energy loss	**B1**	
		(ii)	rms output $V = 24 / \sqrt{2} = 16.97$ V **or** peak input $V = 240 \times \sqrt{2} = 339.4$ V $N_p / N_s = V_p / V_s$ (V_p and V_s either **both** rms or **both** peak) $N_p = 240 / 16.97 \times 260$ **or** $339.4 / 24 \times 260 = 3680$ **or** 3700 (3 or 2 sf)	**C1** **C1** **A1**	
		(iii)	mean input power = 0.35 × 240 = 84 W mean output power = 3.5 × 16.97 = 59.40 W Calculate input and output powers (either **both** peak or **both** mean) efficiency = (output power / input power) × 100% = 70.7 % or 71 %	 **M1** **A1**	
		(iv)	use soft iron as the core to reduce power loss due to repeated magnetisation (and demagnetisation) of the iron core (**or** <u>hysteresis</u> loss) **or** laminate the iron core to reduce power loss due to eddy currents induced in the core	**M1** **A1** **(M1)** **(A1)**	

7	**(a)**		